

20-06-2026

Agenda:

ML-9

- logistic regression •
- practicals
 - ↳ column transformer
 - ↳ pipeline
- AUC curve (evaluation metric)
(manually)

prerequisite:

- confusion matrix
- accuracy
- TP, TN, FP, FN
- f1 score

logistic regression

(1) linear eqn: $z = wx + b$ •
 $= mx + b$ } same

(2) sigmoid prediction: $\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$

(3) log loss function: $-[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$
(J)

Goal of derivative of loss function: → gradient

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w}$$

→ chain rule

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial b}$$

$$\begin{aligned}
 \text{step: } \frac{\partial J}{\partial \hat{y}} &= - \left[y \log(\hat{y}) + (1-y) (\log(1-\hat{y})) \right] \\
 &= - \left(y \cdot \frac{1}{\hat{y}} \cdot 1 + (1-y) \cdot \frac{1}{1-\hat{y}} \cdot (-1) \right) \\
 &= - \left(\frac{y}{\hat{y}} - \frac{(1-y)}{(1-\hat{y})} \right) \\
 &= - \left(\frac{y(1-\hat{y}) - \hat{y}(1-y)}{\hat{y}(1-\hat{y})} \right) \\
 &= - \left(\frac{\cancel{y} - \cancel{y}\hat{y} - \hat{y} + \cancel{y}\hat{y}}{\hat{y}(1-\hat{y})} \right) = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}
 \end{aligned}$$

$$\begin{aligned}
 \text{step 2: } \frac{\partial \hat{y}}{\partial z} &= \hat{y} \cdot (1-\hat{y}) \\
 &\text{(sigmoid)}
 \end{aligned}$$

step 3 derivative w.r.t w
 derivative w.r.t b

$$\frac{\partial z}{\partial w} = x \qquad \frac{\partial z}{\partial b} = 1$$

$$\begin{aligned}\frac{\partial J}{\partial w} &= \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w} \\ &= \left[\frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \right] \cdot [\hat{y}(1-\hat{y})] \cdot [x] \\ &= (\hat{y} - y) x\end{aligned}$$

$$\begin{aligned}\frac{\partial J}{\partial b} &= \frac{\partial J}{\partial \hat{y}} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial b} \\ &= \left[\frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \right] \cdot [\hat{y}(1-\hat{y})] \cdot [1] \\ &= (\hat{y} - y)\end{aligned}$$

$$w_{\text{new}} = w_{\text{old}} - \alpha \cdot \text{gradient} \rightarrow \text{w.r.t} \rightarrow w$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \cdot \text{gradient} \rightarrow \text{w.r.t} \rightarrow b$$

One iteration of Logistic Regression (Dummy Data)

Tumor-size (x)	malignant (y)	}	Dataset
2	1		

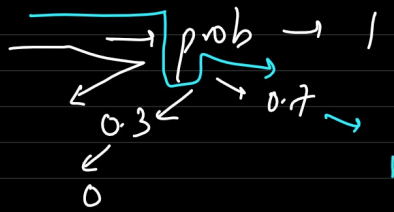
→ Initialize random w & b :

$$\begin{aligned}w &\rightarrow 0.5 \\ b &\rightarrow 0\end{aligned}$$

$$\alpha = 0.1 \text{ (learning rate)}$$

Step 1:

$$z = w \cdot x + b$$
$$= 0.5(2) + 0 \rightarrow 1.0$$



Step 2:

$$\hat{y} = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1.0}} \approx 0.7310586$$

→ **current state:** The model predicts a 73.1% chance of malignancy

Step 3: Calculate the gradient

$$\frac{\partial J}{\partial w} = (\hat{y} - y) \cdot x = (0.731 - 1) \cdot 2$$
$$= -0.538$$

$$\frac{\partial J}{\partial b} = (\hat{y} - y) \cdot 1 = -0.269$$

Step 4: update w & b

$$w_{\text{new}} = w_{\text{old}} - (\alpha \cdot \text{gradient}_w)$$
$$= 0.5 - (0.1 \cdot -0.538)$$
$$= 0.5 - (-0.0538)$$
$$= 0.5538$$

$$b_{\text{new}} = b_{\text{old}} - (\alpha \cdot \text{gradient}_b)$$
$$= 0 - (0.1 \cdot -0.269)$$
$$= 0.0269$$

step 5: Error Reduced?

$$\text{Original output / prob} = 0.731$$

$$\text{new output / prob} = z = (0.5538 \cdot 2) + 0.0269 = 1.1345$$

$$\hat{y} = \text{sigmoid}(z) \approx 0.7567$$

prediction moved from 0.731 to 0.756

→ closer towards 1